MILITARY SPECIFICATION

COUPLINGS, HOSE, REATTACHABLE SCREW-ON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers screw-on type reattachable couplings for hoses conforming to MIL-H-6615, MIL-H-26521, as specified (see 6.2).

1.2 Classification. The couplings shall be of the following classes and types, as specified (see 6.2):

Class 1	Fuel service
Type I	External NPT thread
Type II	Internal NPSH thread with swivel
Class 2	Demineralized water service
Type I	External NPT thread
Type II	Internal NPSH thread with swivel

* 1.3 Part numbering. Definitive part numbers shall be in accordance with 6.3.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

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SPECIFICATION

FEDERAL

QQ-P-416	Plating, Cadium (Electrodeposited)
ZZ-H-466	Hose Assembly, Metal (Rubber Covered), Hydrocarbon Conveying.

MILITARY

MIL-H-775	Hose, Hose Assemblies, Rubber, Plastic, Fabric or Metal (Including Tubing); and Fittings, Nozzles and Strainers.
MIL-H-6615	Hose Assemblies, Rubber, Fuel and Water, with Reattachable Couplings, Low Temperature.
MIL-R-6855	Rubber, Synthetic, Sheets, Strips Molded or Extruded Shapes.
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-H-26521	Hose and Hose Assembly, Nonmetallic, Fuel, Collapsible, Low Temperature with Reattachable Couplings.

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

DOD-STD-100	Engineering Drawing Practices.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of.
MIL-STD-810	Environmental Test Methods .
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-1188	Commercial Packaging of Supplies and Equipment.

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Services

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402).

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AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI H35.1 Alloy and Temper Designation System for Wrought Aluminum.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see sections 4 and 6).

3.2 Components. The coupling shall consist of the following major components:

Item name

Body (see 3.7.1)

Grip (see 3.7.2)

Sleeve (see 3.7.3)

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service and shall be entirely suitable for the purpose. Materials shall be free from all defects and imperfections that might affect the serviceability of the finished coupling.

3.4.1 Material for class 1 couplings. The material for class 1 couplings shall be as follows:

Body and sleeve	High-grade brass composition, containing not less than 58 percent copper.
Grip	Steel wire, cadmium plated in accordance with QQ-P-416, type II, class 2.

3.4.2 Material for class 2 couplings. The material for class 2 couplings shall be as follows:

Body and sleeve	300 series stainless steel, type B214 or 356 aluminum casting alloy, or aluminum alloys other than 2000, 4000, and 7000 series in accordance with ANSI H35.1. Aluminum alloys shall be anodized in accordance with MIL-A-8625. The color shall be olive drab, color number 24087 of FED-STD-595.

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Grip	Steel wire, cadmium plated in accordance with QQ-P-416, type II, class 2.

3.5 Design and construction. The couplings shall be designed and constructed for use with hoses conforming to MIL-H-6615 or MIL-H-26521. They shall be reattachable, shall incorporate the feature specified herein and as shown on figure 1, and shall require no cements, sealants, or adhesives in their installation, other than to seal the cut ends of the hose.

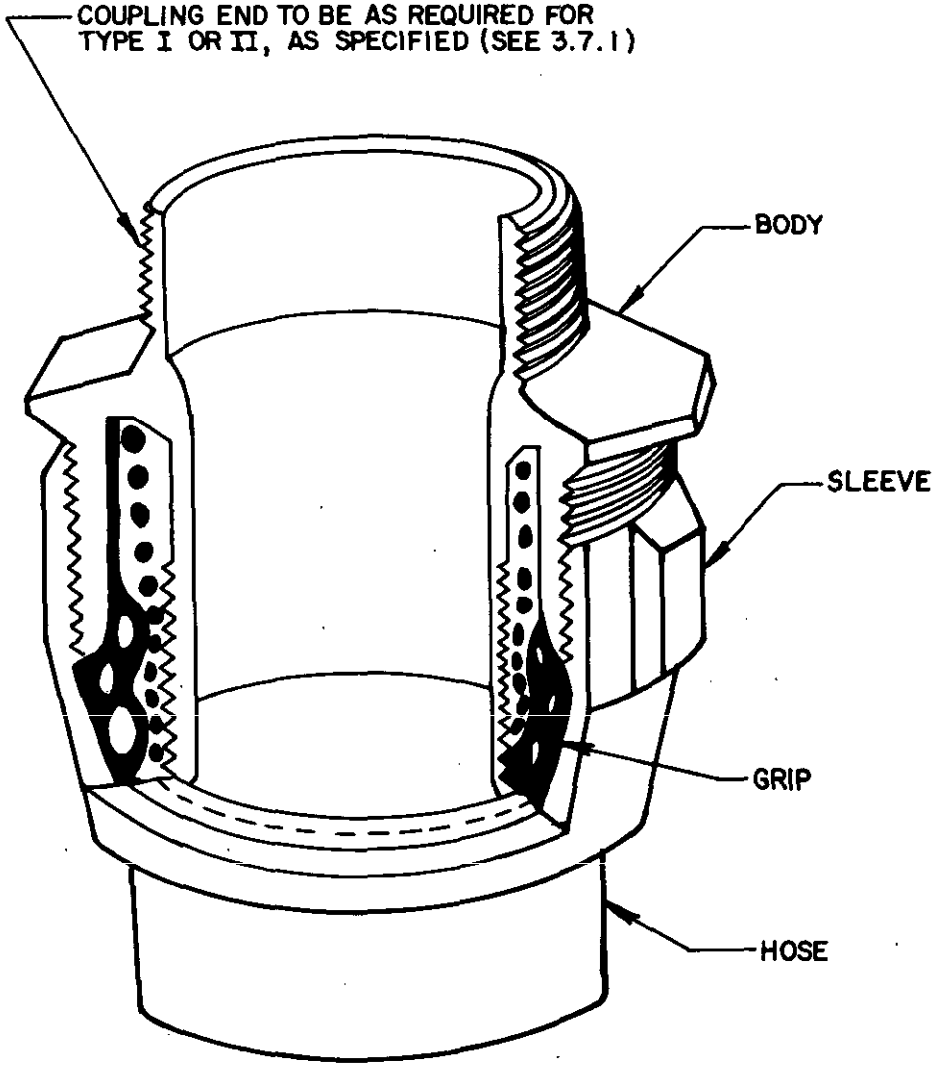


FIGURE 1. Section showing typical coupling assembly construction

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3.5.1 Thread gages. All threads shall be in accordance with Handbook H28. If the male tailpiece threads and female sleeve threads are in accordance with the Handbook H28 special threads (NS or UNS), thread gages shall be provided by the manufacturer. Data specifying the accuracy of the thread gages shall be made available to the procuring activity. (See 3.7.1.1, 3.7.1.2, 3.7.1.3, and 3.7.3).

3.6 Performance.

3.6.1 Environmental conditions. The couplings shall exhibit no corrosion after having been exposed to the following environmental conditions:

a. Relative humidity up to 100 percent, including conditions wherein condensation takes place in the form of water and frost.

b. Salt atmosphere as encountered near bodies of salt water and in transportation on shipboard.

3.6.2 Hydrostatic pressure rating. The internal hydrostatic pressure rating shall be as specified in Table I. The size of the coupling shall be as specified by the procuring activity (see 6.2).

* 3.6.3 Hose retention. The couplings shall be installed on the hose and shall not be detached from the hose when subjected to a repeated longitudinal load. After the repeated load test, couplings shall not blow off or leak when test pressure specified in Table I.

TABLE I. Pressure rating.

Size	Size code	Working pressure (psi)	Test pressure (psi)
1.25	A	400	600
1.50	B	400	600
2	C	380	570
2.50	D	360	540
3	E	320	480
4	F	280	420

3.7 Detail of components.

3.7.1 Body. The body of the coupling shall have a configuration as shown on Figure 2. The body shall have a coupling end for attachment to mating assemblies and a tailpiece end for attachment of the applicable hose. The central hexagon or octogon shall have a larger dimension across the flats than the largest diameter across corners of the coupling sleeve.

3.7.1.1 Type I coupling end. The type I coupling end shall have external American Standard taper pipe threads (NPT). The end surface of the coupling end shall be a smooth finish with rounded edges, square with the thread axis, to provide a good sealing surface when joined to a type II coupling.

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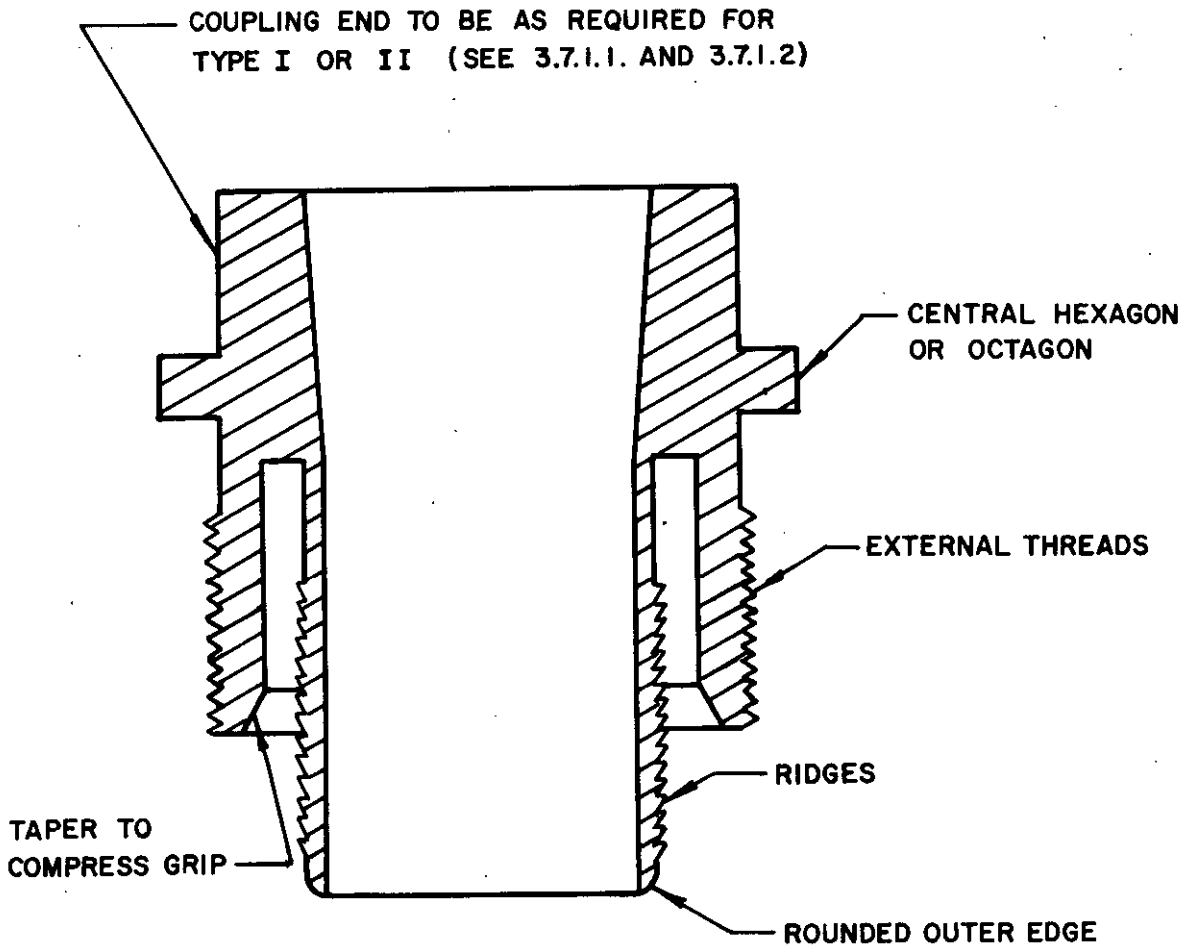


FIGURE 2. Typical cross-section of coupling body

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3.7.1.2 Type II coupling end. The type II coupling end shall have internal American National hose coupling threads (NPSH) which shall swivel on the remainder of the body. The 2 1/2-, 3-, and 4 inch nominal size type II couplings shall conform to the 8 threads per inch sizes of ZZ-H-466. The body shall provide a smooth sealing surface for a gasket. Each body shall be furnished with a gasket of synthetic rubber conforming to MIL-R-6855 class 1. The swivel end shall turn freely by hand, without the use of tools or mechanical assistance, until slightly before a leaktight seal is effected against the gasket.

3.7.1.3 Tailpiece end. The tailpiece end shall consist of two concentric walls spaced to accept hose in accordance with MIL-H-6615 or MIL-H-26521, of the appropriate size. The outer wall of the tailpiece end shall have external threads to engage the sleeve, and the inside end shall be tapered to compress the grip. The external threads shall be straight and no coarser than 10 threads per inch. The inner wall shall have ridges to assure a tight grip on the hose and shall have a rounded outer edge to prevent cutting of the hose during installation or upon sharp bending.

3.7.2 Grip. The grip shall be of material as specified in 3.4.1 and 3.4.2. The wire shall be left-hand wound into a coil of approximately three loops, and may be round or flat with rounded edges. The end of the grip shall be ground to prevent damaging the hose or injuring using personnel during installation or removal of the coupling.

3.7.3 Sleeve. The sleeve shall have a configuration incorporating the features shown on Figure 3 and as specified herein. Four flat surfaces (spaced 90 degrees apart) shall be provided for attaching a wrench. The interior of the sleeve shall have internal threads for connection with those provided on the body, shall be tapered to compress the grip, and shall have the edge rounded to prevent cutting of the hose upon sharp bending. When fully threaded to the body, the sleeve shall extend beyond the body tailpiece.

3.8 Interchangeability. All functionally and dimensionally interchangeable part shall have the same manufacturer's part number. The drawing number requirements of DOD-STD-100 shall govern changes in the manufacturer's part numbers.

* 3.9 Marking. The sleeve specified in 3.7.3 shall be marked (stamped) with the applicable hose size and manufacturer's code in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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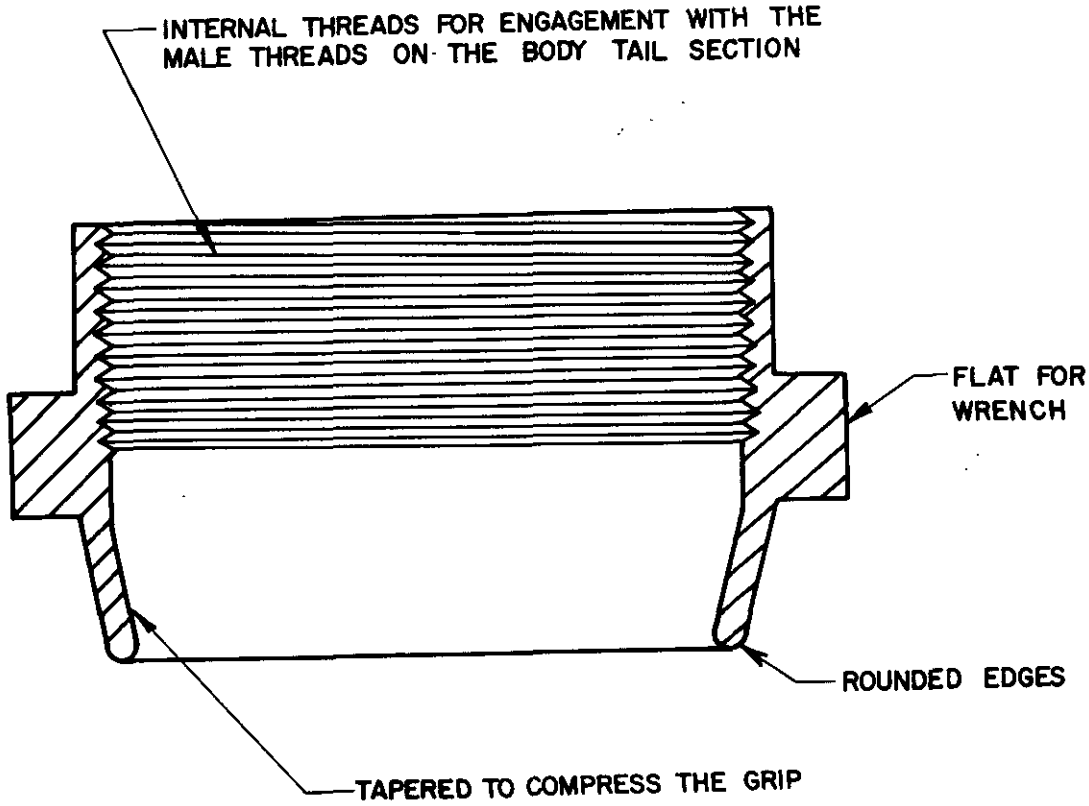


FIGURE 3. Cross-section of coupling sleeve

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4.2 Classification of inspection. The examination and testing of the couplings shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection (see 6.2).

4.3.1 Test samples. One coupling of each class and type being procured for each applicable hose specification shall be subjected to the tests specified in 4.3.3.

4.3.2 Test report. Upon completion of the first article tests, three completed copies of a test report in accordance with MIL-STD-831 shall be supplied to the procuring activity.

4.3.3 First article inspection. The first article inspection shall consist of all the tests described under 4.5.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of the individual tests.

4.4.1 Individual tests. Each coupling shall be subjected to the examination specified in 4.5.1. If the body is cast, each coupling shall be subjected to the test specified in 4.5.2.

4.5 Inspection methods.

4.5.1 Examination of product. The coupling shall be inspected to determine compliance with the requirements specified herein. The following shall be applicable to the first article item only and shall be included in the test report:

a. Weight of component.

b. Dimensions of components.

c. Certification that the materials used complied with the requirements specified herein.

4.5.2 Pressure test. The coupling body shall be subjected to the applicable internal hydrostatic test pressure specified in Table I for not less than 1 minute. Evidence of leakage shall be cause for rejection.

4.5.3 Swivel test. When applicable, the internally threaded end connection of the body shall demonstrate the ability to swivel by hand relative to the tailpiece which is attached to the hose. After this demonstration, the coupling assembly shall be subjected to the test specified in 4.5.4.

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* 4.5.4 Hose retention test. The coupling assembly shall be attached to a six foot long section of hose conforming to MIL-H-6615. The hose assembly shall be hung vertically by any convenient means and be used to lift a dead weight of 450 pounds at least one inch off the floor and then lowered. This cycle shall be repeated 25 times. The assembly shall then be subjected to the test pressure specified in Table I for a minimum of one minute. Any evidence of leakage, pull or blow off of the coupling shall be cause for rejection. The above shall be repeated with a six foot long section of hose conforming to MIL-H-26521.

4.5.5 Environmental tests. The coupling shall be tested in accordance with 4.5.5.1 through 4.5.5.3.

4.5.5.1 Humidity. The coupling shall be subjected to humidity in accordance with Procedure I, Method 507, of MIL-STD-810.

4.5.5.2 Salt spray. The coupling shall be subjected to salt spray in accordance with Procedure I, Method 509, of MIL-STD-810.

4.5.5.3 Verification with substantiating data that materials used are not adversely affected by humidity or corrosion may be provided in lieu of the humidity and salt spray testing.

4.6 Packaging, packing, and marking. Preparation for delivery shall be inspected for conformance to section 5.

5. PACKAGING

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with applicable levels of MIL-H-775 or MIL-STD-1188, as specified (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The couplings covered by this specification are intended for use on hose conforming to MIL-H-6615 or MIL-H-26521 in ground fueling of aircraft with hydrocarbon fuels (use class 1) or demineralized water (use class 2).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Class, type and size (see 1.2, 3.5, 3.6.2, 3.7.1, and figure 1).

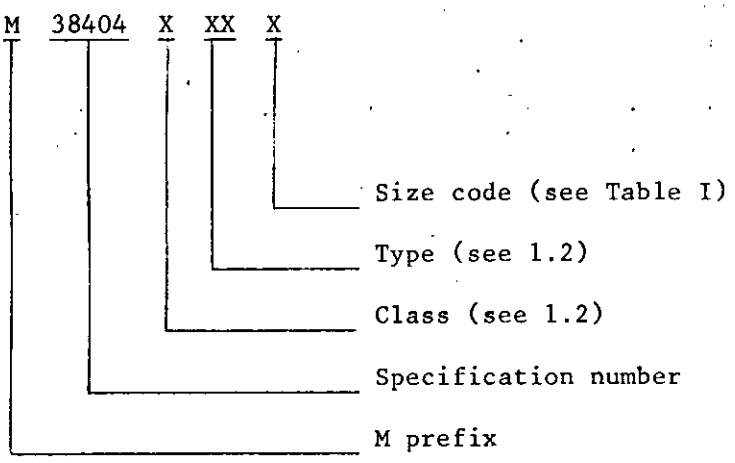
c. Location and conditions for first article inspection (see 4.3).

d. Applicable levels of preservation, packaging, and packing required (see section 5).

e. Special shipment marking (see 5.2).

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6.3 Part number. Definitive part number shall be in accordance with the following:



Example:

M 38404-2-II-C

Class 2, type II, 2 inch I.D.

* 6.4 Changes from previous issue. The margins of this document are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractor's are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-38404D	Couplings, Hose, Reattachable Screw-On

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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